CareLoop

Post-Discharge Recovery & Patient Safety Guardian

DSH HACKS V2 · AI X HEALTHCARE

Team CareLoop · Oct 2026

1. The Critical Clinical Problem

The days immediately following hospital discharge represent the highest-risk vulnerability window in the continuum of care.

- **Discharge Summary Disconnect:** Summaries are dense, abbreviated medical English. Patients and family caregivers leave confused about medication changes and red flags.
- **Medication Hazards:** Complex regimen changes frequently cause accidental double-dosing, missed critical doses, and severe OTC drug-drug interactions.
- **Silent Deterioration:** Congestive fluid overload or postoperative infection begins days before an emergency readmission, with no remote visibility.

2. The Solution: Four Continuous Care Loops

Loop 1 · DECODE (Discharge AI Parser)
Ingests unstructured clinical notes and maps to plain language across 6 Global Languages (English, Español, 中文, Français, Melayu, தமிழ்) with dual reading modes (Simple Everyday ↔ Standard Clinical ICD-10) and source-span highlighting.

Loop 2 · PROTECT (Medication Guardian)
Reconciles active prescriptions against OTC items. Detects fatal drug-drug interactions (e.g. NSAID + Blood Thinner, Triple Whammy acute renal failure) before ingestion via a 32-rule interaction matrix.

Loop 3 · WATCH (60s Check-in & Rules Engine)
Tracks weight, vitals, and symptoms. Computes an explainable 0–100 Recovery Risk Index with transparent factor attribution (Green / Amber / Red). Time Machine follows timeline progression from Day 0 to Day 3.

Loop 4 · CONNECT (Caregiver Pass & Clinician SBAR)
Provides a cryptographically signed offline QR pass (/pass/token) for instant, secure family surveillance, dispatches real-time alerts, and auto-generates a one-tap standardized SBAR brief for outpatient doctor reviews.

3. Verified Clinical Evidence

- **1 in 5 Medicare Patients Readmitted (19.6%):** Jencks SF, Williams MV, Coleman EA. *Rehospitalizations among patients in the Medicare fee-for-service program.* N Engl J Med. 2009;360(14):1418-1428.
- **50% Medication Discrepancies Post-Discharge:** Kripalani S et al. *Deficits in communication and information transfer between hospital and primary care physicians.* JAMA. 2007;297(8):831-841.
- **Preventable Heart Failure Readmissions:** Ponikowski P et al. *2016 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure.* Eur Heart J. 2016;37(27):2129-2200.

Safety-First Hybrid Architecture

AI / NLP Layer: Multilingual translation, medical abbreviation decoding, and dual-mode health literacy adaptation.
Deterministic Rules Engine: 100% of clinical safety decisions, risk scores, drug clashes, and emergency flags run in auditable deterministic code. The AI can never lower risk or alter doses.

RECOVERY TIME MACHINE: DAY 3 CATCH — RED ALERT (100/100)

- **Day 0-1:** Green (68.0 kg) · Baseline stable post-discharge recovery
- **Day 2:** Amber (68.9 kg) · Missed diuretic dose flagged by Medication Guardian
- **Day 3: RED (70.4 kg, +2.4 kg in 3 days)** · Orthopnea & ankle swelling detected. System automatically dispatches family alert and prepares Clinician SBAR handover prior to emergency hospitalisation.

4. Key Technical Differentiators

- **Offline-First Architecture:** Runs client-side with zero server dependency, zero API key requirement, and zero protected health information leakage.
- **100% Provenance Traceability:** Every care plan card highlights the exact verbatim source lines in the primary clinical discharge summary.
- **Live Ground-Truth Evaluation:** Executes 50 live automated tests: 20 clinical summaries (100.0% F1) + 30/30 clinical rules engine test cases.
- **Universal Safety Net:** Always-present emergency protocol displays immediate 911 / 112 / 999 global dialer with high-contrast paramedic handover card.

5. Pilot Roadmap & Impact

Phase 1 (Complete): Zero-dep client with 68 drugs, 32 interactions, 6 global languages, dual reading modes, offline QR pass, Time Machine, and SBAR brief.
Phase 2 (Hospital EHR Pilot): SMART-on-FHIR connector to ingest discharge summaries directly from Epic and Cerner EHRs.
Projected Impact: Potential 25–35% reduction in avoidable 30-day readmissions through early outpatient intervention.